Name: _____

WORKSHEET

Stationary points

1 Given the function $f(x) = 3x^2 - 12x + 5$

a Find the derivative $f'(x)$

b Find the x value of any stationary point(s)

c Find the coordinates of any stationary point(s) of $f'(x)$

2 Find the x value of stationary points for the following functions:

a $f(x) = \frac{1}{2}x^2 + 10x - 5$

b $f(t) = 3 - 5t - 2t^2$

c $g(x) = (x - 4)(x + 3)$

d $h(t) = (2t - 1)(t - 3)$

e $f(x) = \frac{1}{2}x^2 + 7x - \frac{1}{3}$

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f $g(x) = x^3 - \frac{11}{2}x^2 + 6x - 5$

g $f(x) = \frac{1}{3}x^3 + \frac{1}{2}x^2 - 6x + 3$

h $f(x) = x^3 + \frac{3}{2}x^2 - 6x + 3$

i $h(x) = x^3 - 12x - 7$

j $f(x) = 2x^3 - 3x^2 - 120x + 7$

k $g(x) = 14 - 9x^2 - x^3$

l $f(x) = x^3 + x^2 - x + 3$

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3 Given $x(t) = t^3 - 2t^2 - 7t + 1$,

a Find stationary points of $x(t)$.

b Determine the type of each stationary point.

4 For the following functions, find and determine the type of each stationary point.

a $f(x) = x^2 + 6x - 2$

b $g(x) = x^2 - 4x + 1$

c $h(x) = x^2 + 10x - 3$

d $f(x) = 3x^2 - 12x + 5$

e $g(x) = -2x^2 + 8x - 3$

f $h(x) = 11 - 6x - x^2$

g $f(x) = (x + 3)(x - 5)$

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h $g(x) = x^3 - 3x^2 + 7$

i $h(x) = x^3 - 12x - 2$

j $f(x) = x^3 - 6x^2 - 15x + 7$

k $g(x) = 2x^3 - 54x + 5$

l $h(x) = 14 + 27x - x^3$

m $f(x) = x^4 - 32x + 3$

n $f(x) = -x^3 + 8$

Answers

1 **a** $f'(x) = 6x - 12$

b $x = 2$

c $(2, -7)$

2 **a** $x = -10$

b $t = -\frac{5}{4}$

c $x = \frac{1}{2}$

d $t = \frac{7}{4}$

e $x = -7$

f $x = 3, \frac{2}{3}$

g $x = 2, -3$

h $x = -1, 2$

i $x = \pm 2$

j $x = -4, 5$

k $x = 0, -6$

l $x = -1, \frac{1}{3}$

3 **a** $(-1, 5)$ and $\left(2\frac{1}{3}, -13\frac{14}{27}\right)$

b Max $(-1, 5)$, Min $\left(2\frac{1}{3}, -13\frac{14}{27}\right)$

4 **a** Min $(-3, -11)$

b Min $(2, -3)$

c Min $(5, 72)$

d Min $(2, -7)$

e Max $(2, 5)$

f Max $(-3, 20)$

g Min $(1, -16)$

h Max $(0, 7)$ and Min $(2, 3)$

i Max $(-2, 14)$ and Min $(2, -18)$

j Max $(-1, 15)$ and Min $(5, -93)$

k Max $(-3, 113)$ and Min $(3, -103)$

l Min $(-3, -40)$ and Max $(3, 68)$

m Min $(2, -45)$

n $(0, -8)$ horizontal point of inflection on a falling curve

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